



“You can’t have energy transition unless you crack the storage problem. Storage has to become viable. All these developed countries that have been talking about climate action have done nothing. The largest quantum of storage set up by them is 450 MW!”

RAJ KUMAR SINGH

UNION MINISTER OF POWER, NEW & RENEWABLE ENERGY



“If you are 85 per cent import-dependent, you have to increase your exploration and production. We are setting up a National Data Repository so that large firms can come and participate in energy exploration in the country’s sedimentary basins”

HARDEEP SINGH PURI

UNION MINISTER OF PETROLEUM AND NATURAL GAS



“India is exhibiting one of the fastest transitions to renewable energy... From launching the International Solar Alliance in 2015 to setting a target of reaching net-zero emissions by 2070, India is on the path of transformation”

PRAVEER SINHA

CEO & MD, TATA POWER

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place policies such as free transmission, open access and non-applicability of cross-subsidy surcharges on renewable energy required for making green hydrogen much in advance.” He is quick to add that India can supply green hydrogen and green ammonia at prices that are less than what other countries can offer.

In the first phase of the mission, the country is going to replace grey hydrogen used in oil refining and grey ammonia used in fertiliser manufacturing for domestic use. India’s current consumption of grey hydrogen and grey ammonia stands at 18 MMT, according to government data. Referring to potential export markets for the commodity, Singh says, “Other countries need green hydrogen and green ammonia but don’t have the sun or land to produce renewable energy. But we have a competitive advantage; we have the sun, land and a thriving green energy

industry.” The government would also be announcing a PLI scheme for electrolyser manufacturing. An electrolyser is used to separate hydrogen and oxygen molecules in water to produce green hydrogen.

Upbeat with India’s initiatives, Ahmad Chatila, Promoter of the US-based venture capital firm Fenice Investment Group and former CEO of Sun Edison, says that with green hydrogen the country has the potential to become a global energy supplier like Saudi Arabia. “In energy transition technologies, you need three things—human capital; land and natural resources like sunlight and wind speed; and capital. India has the first two in abundance. So, India can do 1 to 2 terawatts (TW or 1,000 GW) of wind and hydrogen. No questions there.” Already, the EU member states, Japan and South Korea have evinced interest in sourcing green hydrogen and green ammonia

supplies from India. Green hydrogen and other low-carbon technologies can potentially create an \$80-billion market in India by 2030, says IEA.

But it’s not going to be easy. The country will need on average \$160 billion per year in investments across its energy economy between now and 2030 to meet its zero-emission goals, projects IEA. Another study by Standard Chartered puts this at \$17.77 trillion by 2070. Therefore, arranging access to low-cost long-term capital will be important. Yet, despite the challenges, India’s success will pave the way for other developing economies to also aspire for a low carbon robust economic growth. Or, as Honeywell’s Gaikwad puts it, it will establish India as the global leader in “the energy transition from molecules (fossil fuels) to electrons (renewables)”. **BT**

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